

USER & TECHNICAL GUIDE

Daily Architecture News Agent

An autonomous AI agent that researches architecture & sustainability news and delivers a daily digest via Telegram

Next.js

TypeScript

Gemini AI

Supabase

Telegram Bot API

Vercel

Crontap

Live application · daily-architecture-news-agent.vercel.app

Prepared by Shahd-Amna-Oroba-Malak

1. What This Project Does

The Daily Architecture News Agent is an autonomous AI system. Every day, at a time each user chooses, it researches the latest architecture and sustainability news, summarizes it using Google's Gemini AI, and delivers a clean digest straight to the user's Telegram — no manual searching required.

Each run performs the following, independently for every registered user:

- 1 Reads the user's preferences** — their tracked keywords (e.g. Architecture, Sustainable Design, BIM, Green Building), how many stories they want, and their chosen delivery hour.
- 2 Researches with Gemini AI** — calls the Gemini API with Google Search grounding to find recent, credible news matching those keywords.
- 3 Summarizes the top stories** — Gemini selects and condenses each story into a title, a one-line summary, and a source link.
- 4 Saves & delivers** — stores the report in the database, then sends a formatted digest to the user's Telegram chat.

Reliable by design

Each user is processed in isolation. If one user's delivery fails, it never blocks the others — the run reports per-user success and failure.

2. User Guide — How to Try It

Follow these steps to register and receive your own daily digest. The whole process takes about three minutes.

Step A — Set up the Telegram bot

- 1** Open Telegram and go to the bot: t.me/arch_news_AI_Agent_bot
- 2** Press **START** and send any message (e.g. "hi"). *This is required* — a bot can only message you after you message it first.
- 3** Open [@userinfobot](#) in Telegram and press **START**. It replies with your **Chat ID** — a number (e.g. [6298966365](#)). Copy it.

Step B — Register in the app

- 1** Open the app: daily-architecture-news-agent.vercel.app

- 2 On the setup page, fill in:
 - **Email** — used only to identify your account.
 - **Telegram Chat ID** — the number from `@userinfobot` (your own, not anyone else's).
 - **Number of news items** — e.g. 5.
 - **Keywords** — comma-separated topics to track.
 - **Preferred daily time** — shown in Jordan time; pick the hour you want your digest.
- 3 Press **Save and continue**. You'll be taken to your dashboard.
- 4 To test immediately, press **Run now** on the dashboard — your digest arrives on Telegram within a few seconds.
- 5 From then on, your digest is delivered automatically every day at the hour you chose.

If the app link won't open

Some networks in the region block `vercel.app` domains. If the page won't load, try mobile data instead of Wi-Fi, or use a VPN. The app itself is online and working — this is only a local network restriction.

Important: use your own Chat ID

The Chat ID decides where the message goes. If you enter someone else's number, the digest is delivered to them, not you. Always use the number that `@userinfobot` gives you.

3. Technical Overview — How It's Built

This section explains the architecture and technology choices for anyone reviewing how the system works under the hood.

Technology stack

Layer	Technology	Role
Frontend & API	Next.js (TypeScript)	Setup form, dashboard, and serverless API routes
AI research	Google Gemini API	Finds and summarizes news with Google Search grounding
Database	Supabase (PostgreSQL)	Stores user configs and generated reports
Delivery	Telegram Bot API	Sends the formatted digest to each user
Hosting	Vercel	Deploys the app; auto-redeploys on every Git push
Scheduling	Crontap (external cron)	Calls the agent hourly so per-user times work on the free tier

Data model (Supabase)

Table	Purpose
<code>user_config</code>	One row per user: email, keywords, number of items, chosen run time (stored in UTC), and Telegram chat ID.
<code>daily_report</code>	A record of each generated digest for history and auditing.

How automatic delivery works

Vercel's free (Hobby) plan only allows a cron job to run once per day, which can't support each user picking their own hour. The solution is an external scheduler:

- 1 Crontap calls the agent every hour** — it sends an authenticated request to the `/api/cron` endpoint, once per hour, around the clock.
- 2 The endpoint checks the current UTC hour** — and finds only the users whose chosen time matches this hour.
- 3 It responds instantly** — returning `{ status: "started" }` in under a second, then runs the Gemini + Telegram work in the background (via Vercel's `waitUntil`) so the scheduler never times out.

Security

The `/api/cron` endpoint is protected by a secret token (`CRON_SECRET`). Every scheduled call must include it as an `Authorization: Bearer` header; requests without the correct token are rejected with **401 Unauthorized**. All API keys (Gemini, Supabase, Telegram) are stored as environment variables in Vercel and are never committed to the public code repository.

Time handling

Users pick their delivery time in Jordan time (Amman, permanently UTC+3, with no daylight saving). The setup form converts this to UTC before saving, and the scheduler compares against the current UTC hour — keeping the backend consistent while showing users a familiar local time.

Deployment workflow

The code lives on GitHub. Vercel is connected to the repository, so every `git push` to the main branch automatically builds and deploys a new version to the live URL — with free HTTPS.

Cost

The entire system runs on free tiers — Vercel, Supabase, Gemini, Telegram, and Crontap — with no paid subscription required.